| Feature | B-DNA | A-DNA | Z-DNA |
|---|---|---|---|
| Type of helix | Right-handed | Right-handed | Left-handed |
| Helical diameter (nm) | 2.37 | 2.55 | 1.84 |
| Rise per base pair (nm) | 0.34 | 0.29 | 0.37 |
| Distance per complete turn (pitch) (nm) | 3.4 | 3.2 | 4.5 |
| Number of base pairs per complete turn | 10 | 11 | 12 |
| Topology of major groove | Wide, deep | Narrow, deep | Flat |
| Topology of minor groove | Narrow, shallow | Broad, shallow | Narrow, deep |

coding, decoding, regulation and expression of genes. RNA is single stranded and is unstable when compared to DNA.

### 8.7.5 Types of RNA

- **mRNA (messenger RNA):** Single stranded, carries a copy of instructions for assembling amino acids into proteins. It is very unstable and comprises 5% of total RNA polymer. Prokaryotic mRNA (Polycistronic) carry coding sequences for many polypeptides. Eukaryotic mRNA (Monocistronic) contains information for only one polypeptide.
- **tRNA (transfer RNA):** Translates the code from mRNA and transfers amino acids to the ribosome to build proteins. It is highly folded into an elaborate 3D structure and comprises about 15% of total RNA. It is also called as **soluble RNA**.
- **rRNA (ribosomal RNA):** Single stranded, metabolically stable, make up the two subunits of ribosomes. It constitutes 80% of the total RNA. It is a polymer with varied length from 120–3000 nucleotides and gives ribosomes their shape. Genes for rRNA are highly conserved and employed for phylogenetic studies (Figure 8.28).

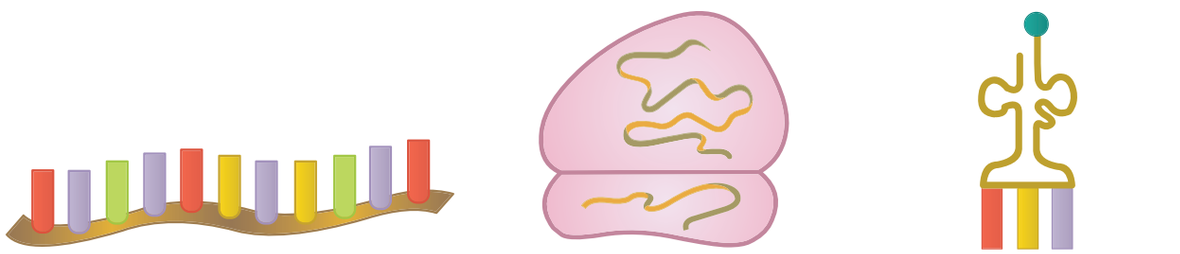



**Figure 8.28:** Types of RNA

## Summary

- **Cells** are composed of water, inorganic compounds and organic molecules. The biomolecules of the cells include carbohydrates, lipids, proteins, enzymes and nucleic acids.
- **Carbohydrates** include simple sugars (monosaccharides) and polysaccharides. Polysaccharide serve as storage forms of sugar and structural components of cell.
- **Lipids** are the principle components of cell membrane, and they serve as energy storage and signalling molecules.
- **Proteins** are polymers of 20 different amino acids, each of which has a distinct side chain with specific chemical properties. Each protein has a unique aminoacid sequence which determines its 3D structure.



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